



Lab No: 01

**Lab Title: GitHub Identify, Student Portal Registration, and
Ubuntu Server Installation**

Name: Tuba Zainab

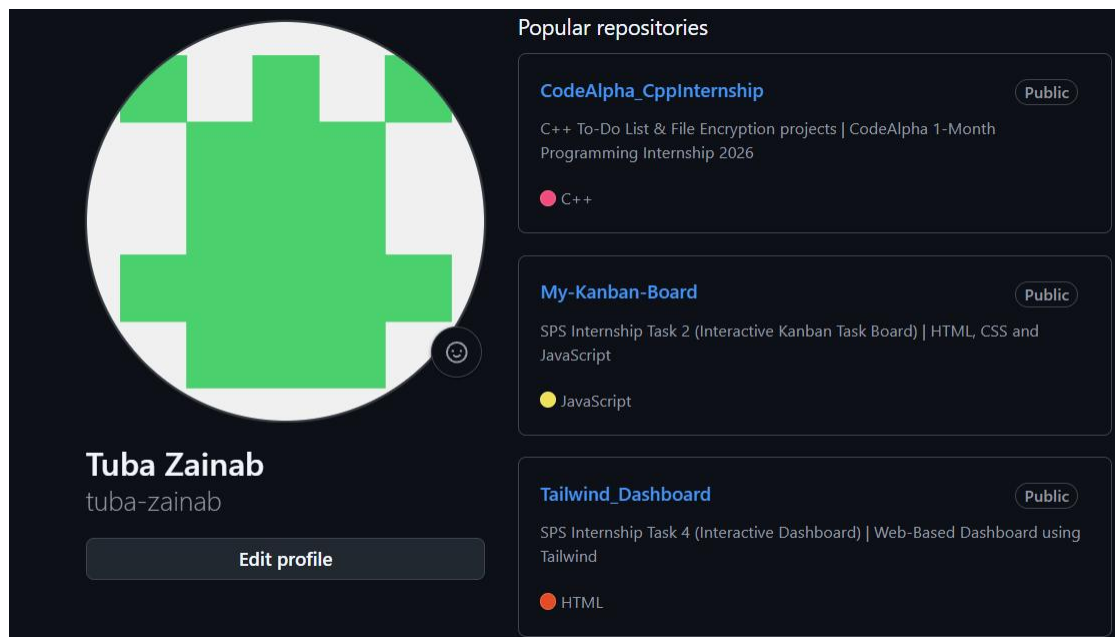
Registration No: FA24B1-SE-006

Course Code: SE-665

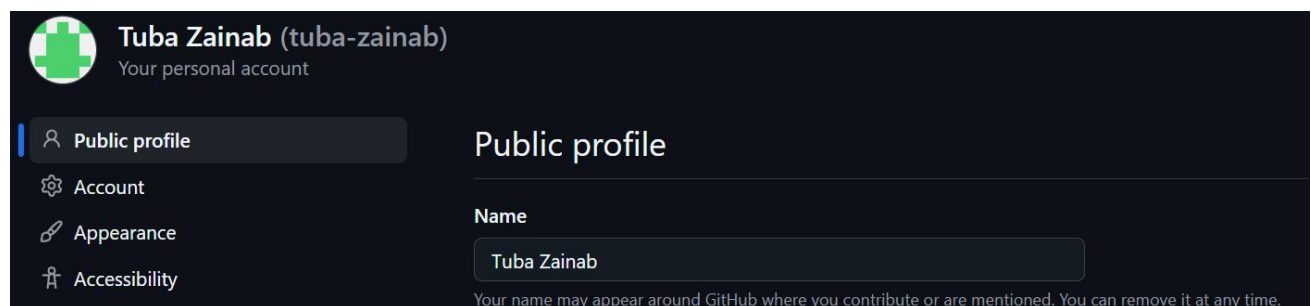
Section: B

GitHub Username: tuba-zainab

Task 01: Create and Configure Your GitHub Account

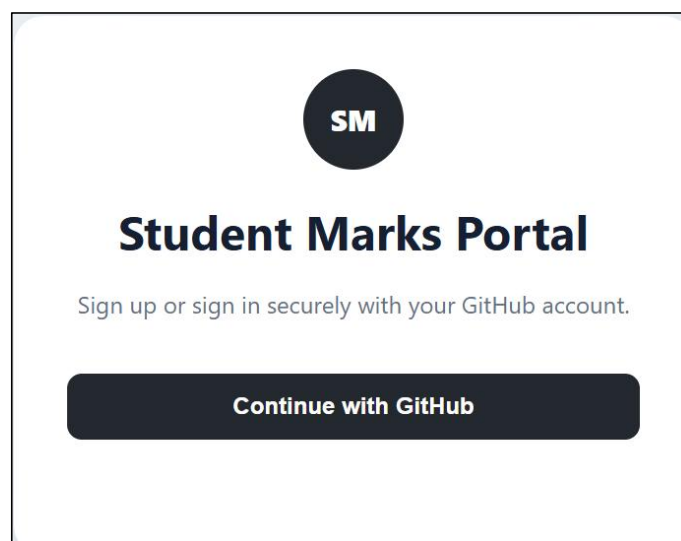


This screenshot clearly shows my GitHub profile name and username.

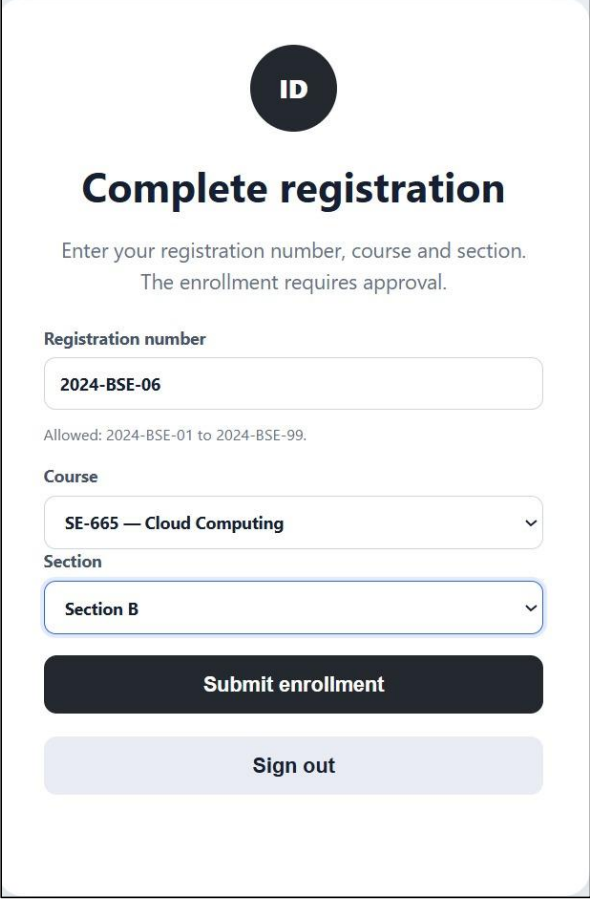


The completed Name field can be seen in this screenshot.

Task 02: Register in the Student Marks Portal

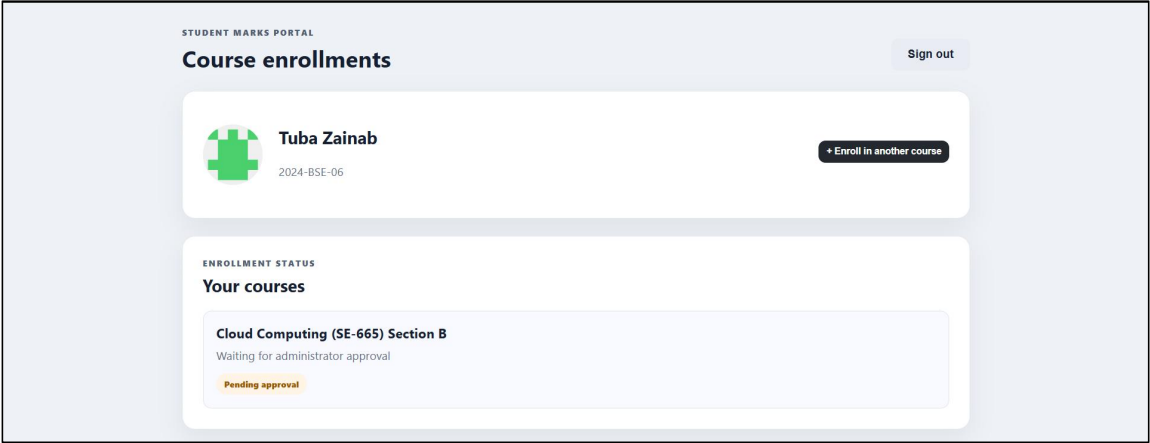


This is the login page of the Student Marks Portal where I will select continue with GitHub in order to access this login portal and link my profile with it.



The screenshot shows a registration form titled "Complete registration". At the top, there is a dark circular button labeled "ID". Below the title, a subtitle reads "Enter your registration number, course and section. The enrollment requires approval." The form contains three input fields: "Registration number" with the value "2024-BSE-06", "Course" with the value "SE-665 — Cloud Computing", and "Section" with the value "Section B". Below these fields are two buttons: a dark "Submit enrollment" button and a light "Sign out" button. A small note below the registration number field states "Allowed: 2024-BSE-01 to 2024-BSE-99."

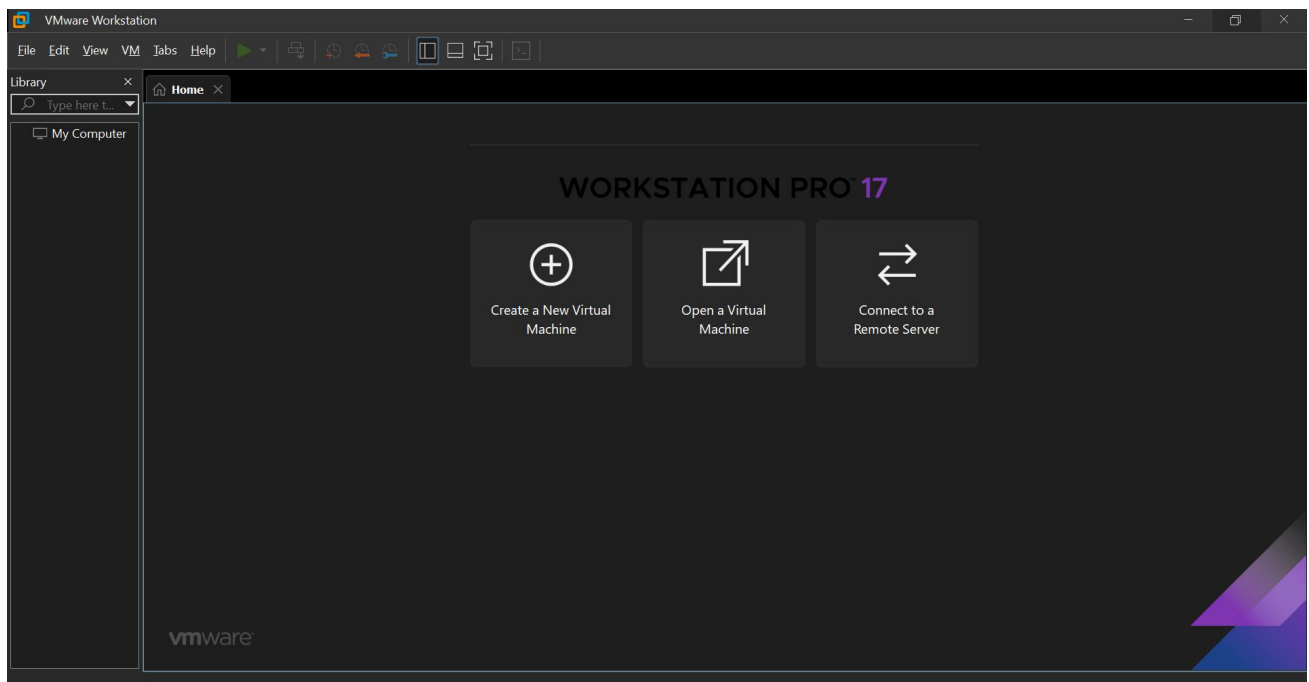
In order to submit the enrollment, I entered the required details. The details can be seen here.



The screenshot shows the "Course enrollments" page. At the top, it says "STUDENT MARKS PORTAL" and "Course enrollments". There is a "Sign out" button in the top right. Below this, a user profile card for "Tuba Zainab" with ID "2024-BSE-06" is shown, featuring a green cross icon and a "+ Enroll in another course" button. Below the profile card, the "ENROLLMENT STATUS" section is titled "Your courses". It lists "Cloud Computing (SE-665) Section B" with a status of "Waiting for administrator approval" and a "Pending approval" label.

This screenshot shows the submitted enrollment for the Student Marks Portal having a pending status.

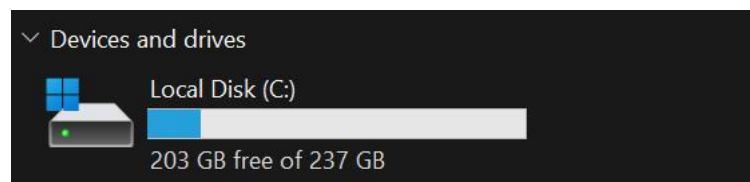
Task 03: Download VMware Workstation Pro and Ubuntu Server



This screen shows that VMware Workstation Pro is installed, it works and is running.

 ubuntu-26.04.1-live-server-amd64	13/09/2026 10:01 am	Disc Image File	2,859,240 KB
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This is the downloaded file of latest Ubuntu Server LTS ISO which is place in my Laptop's Download folder ensuring its existence.



For Virtual Machines, at least 20 GB of storage should be free, which can be seen in his image as available.

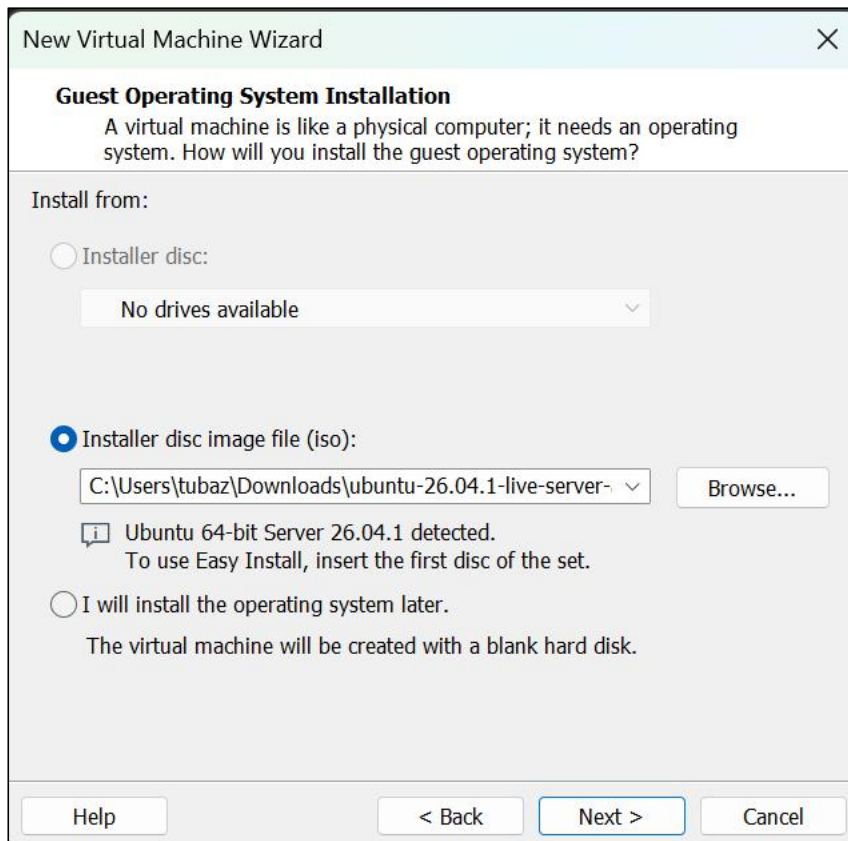
Task 04: Create the Ubuntu Server Virtual Machine



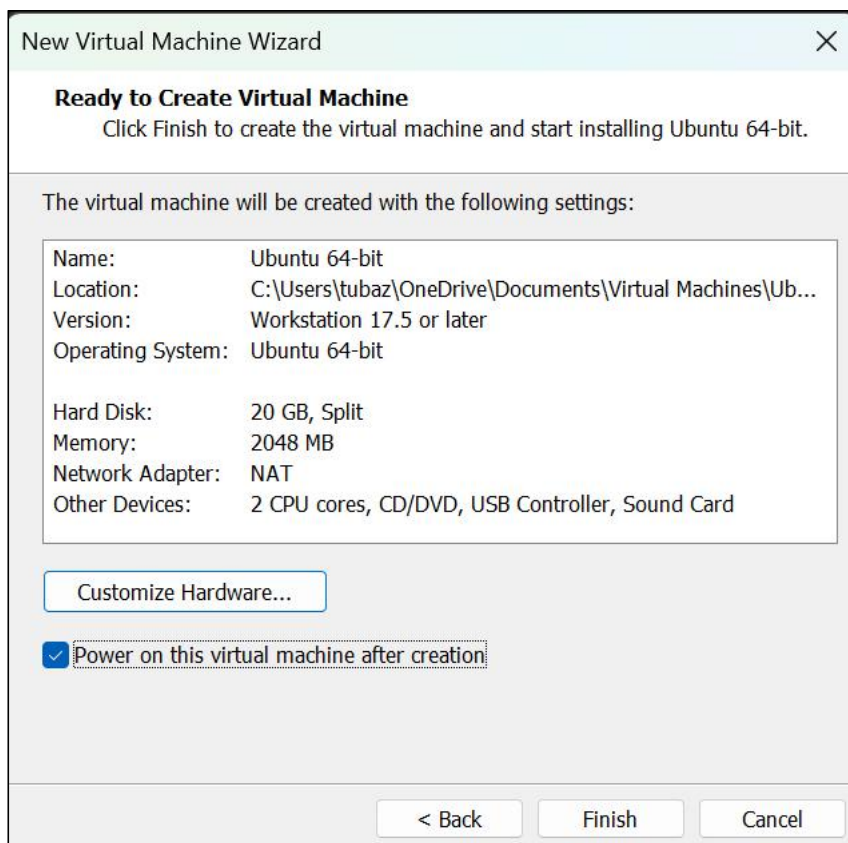
This is the first step after selecting “Create a New Virtual Machine”.



For Virtual Machine Configuration, the recommended option “Typical” is selected here.



In this step, the Installer disc image file (ISO) is selected which was downloaded earlier in Task 3.

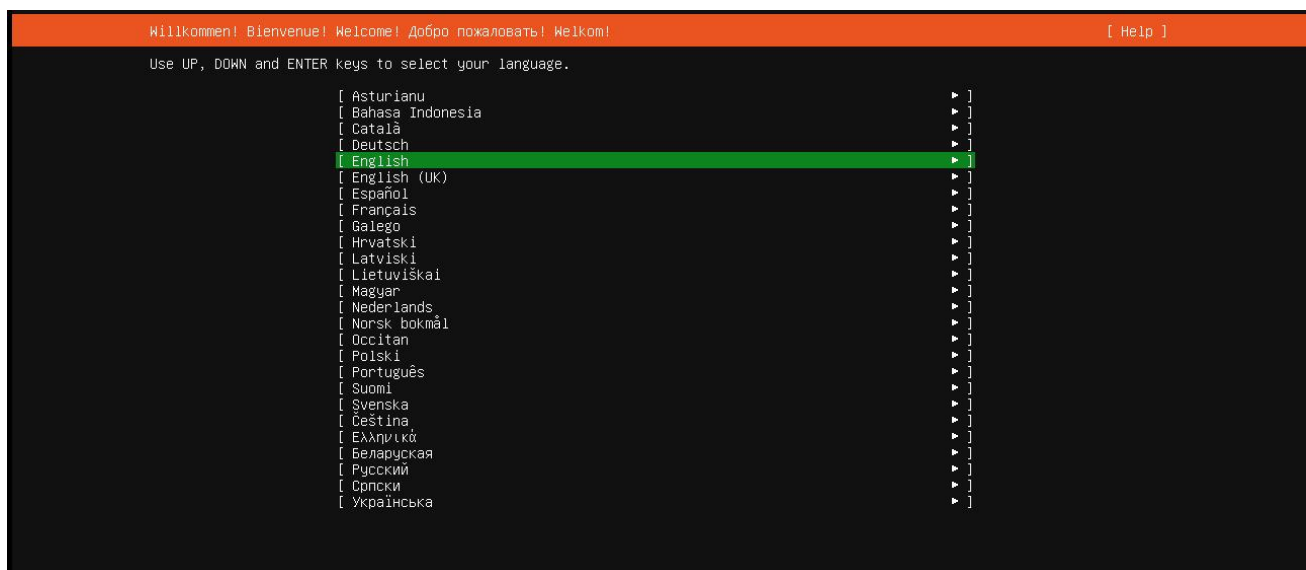


The Virtual Machine Configuration is done in this step by selecting 20 GB virtual disk space and required memory.

```
Starting snapd.apparmor.service - δfiles managed internally by snapd...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalrd.service - Create δruntime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Started systemd-ask-password-console-requests to Console Directory Watch.
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-udev.service - Rebuild Journal Catalog...
[ OK ] Started vgauth.service - Authenticδ virtual machines hosted on VMware.
[ OK ] Started open-vm-tools.service - Setδ virtual machines hosted on VMware.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
Mounting proc-sys-fs-binfmt_misc.mount - Executable File Formats File System...
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Executable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ OK ] Finished systemd-journal-catalog-udev.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
[ 23.894966 ] piix4_smbus 0000:00:07.3: SMBus Host Controller not enabled!
[ OK ] Finished console-setup.service - Set console font and keymap.
[ OK ] Finished snapd.apparmor.service - δfiles managed internally by snapd.
```

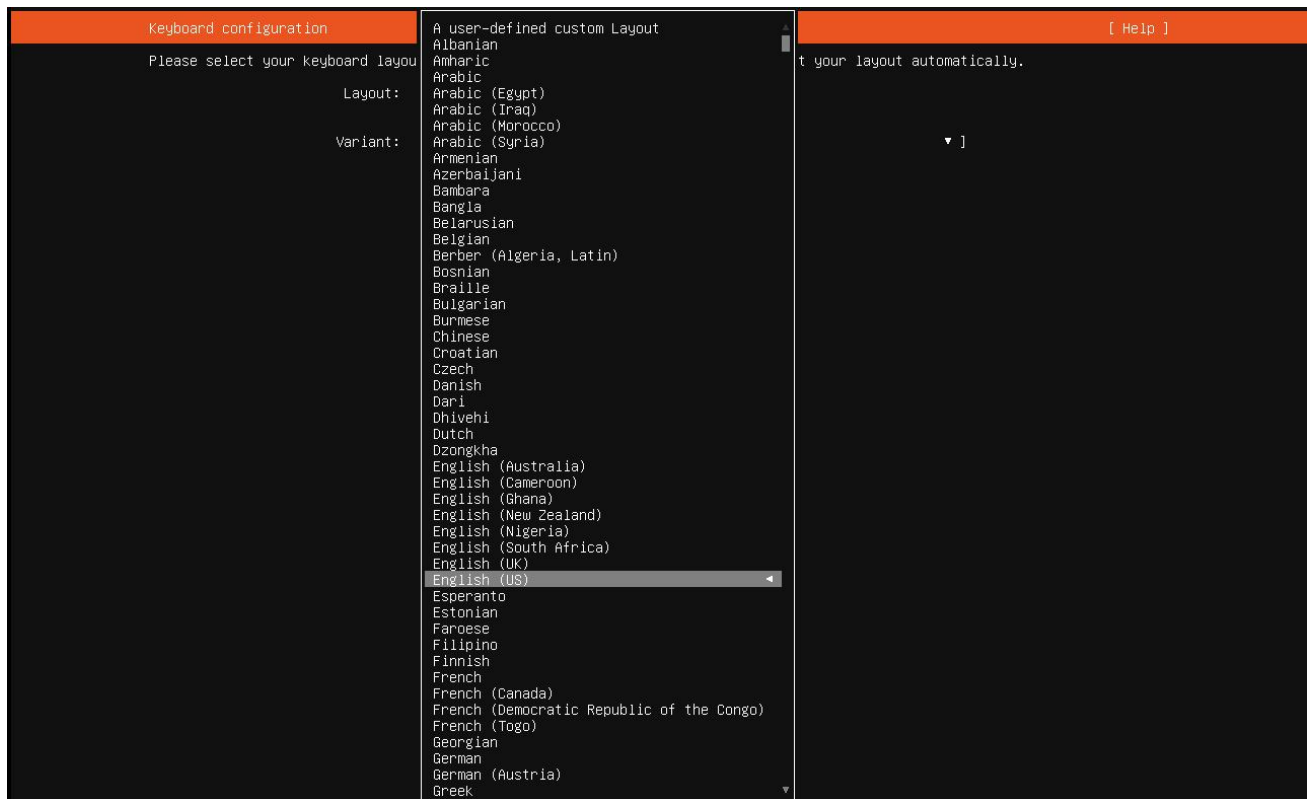
This is the installer screen which started automatically on powering on the Virtual Machine.

Task 05: Choose the Installation Language

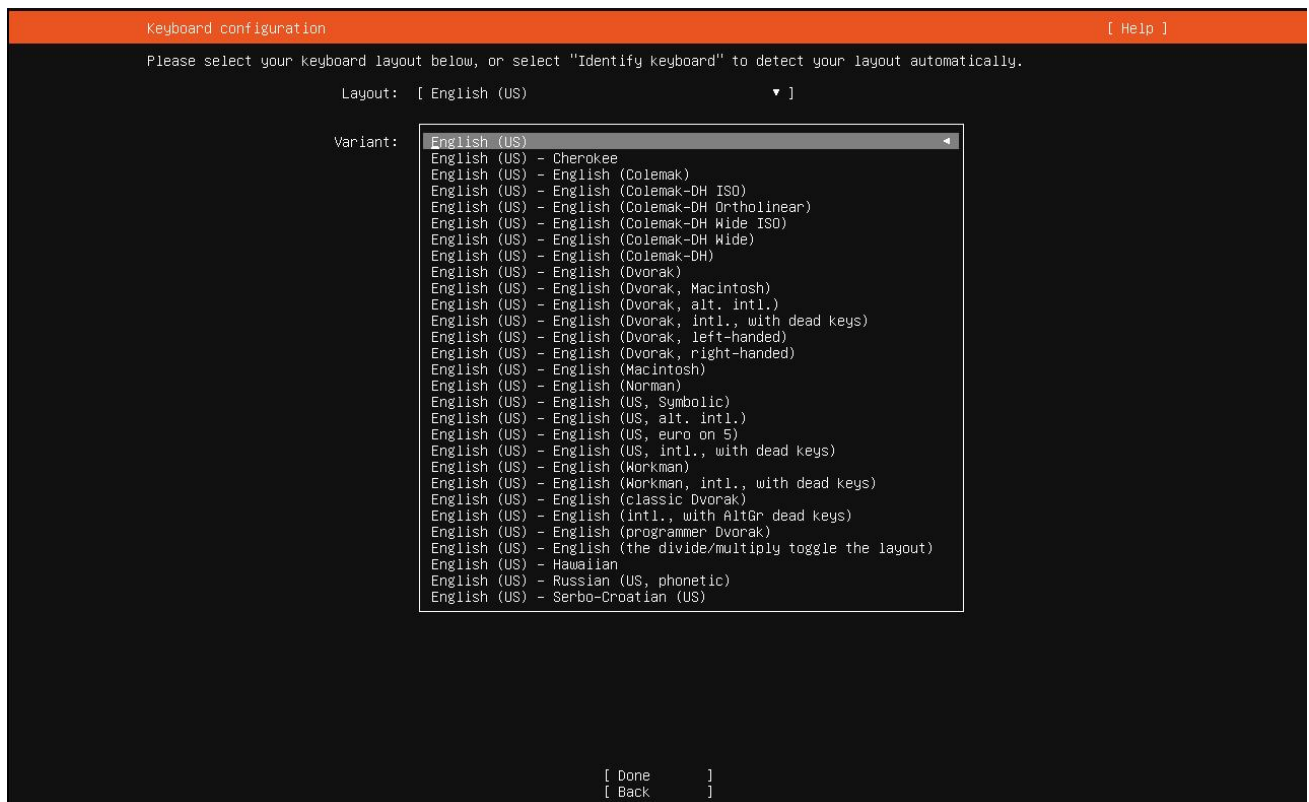


In this step, the preferred language is selected which is English (US).

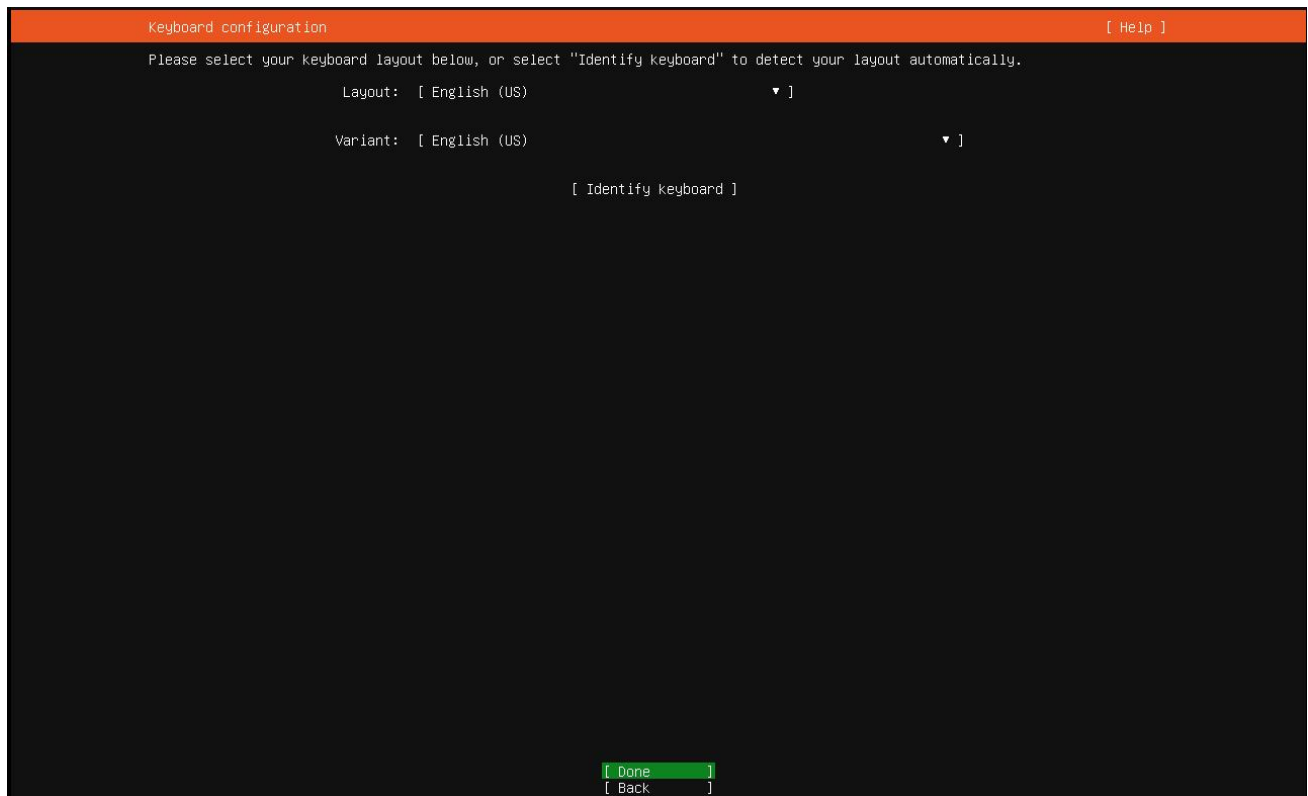
Task 06: Configure the Keyboard Layout



First the keyboard layout is selected which is English (US).

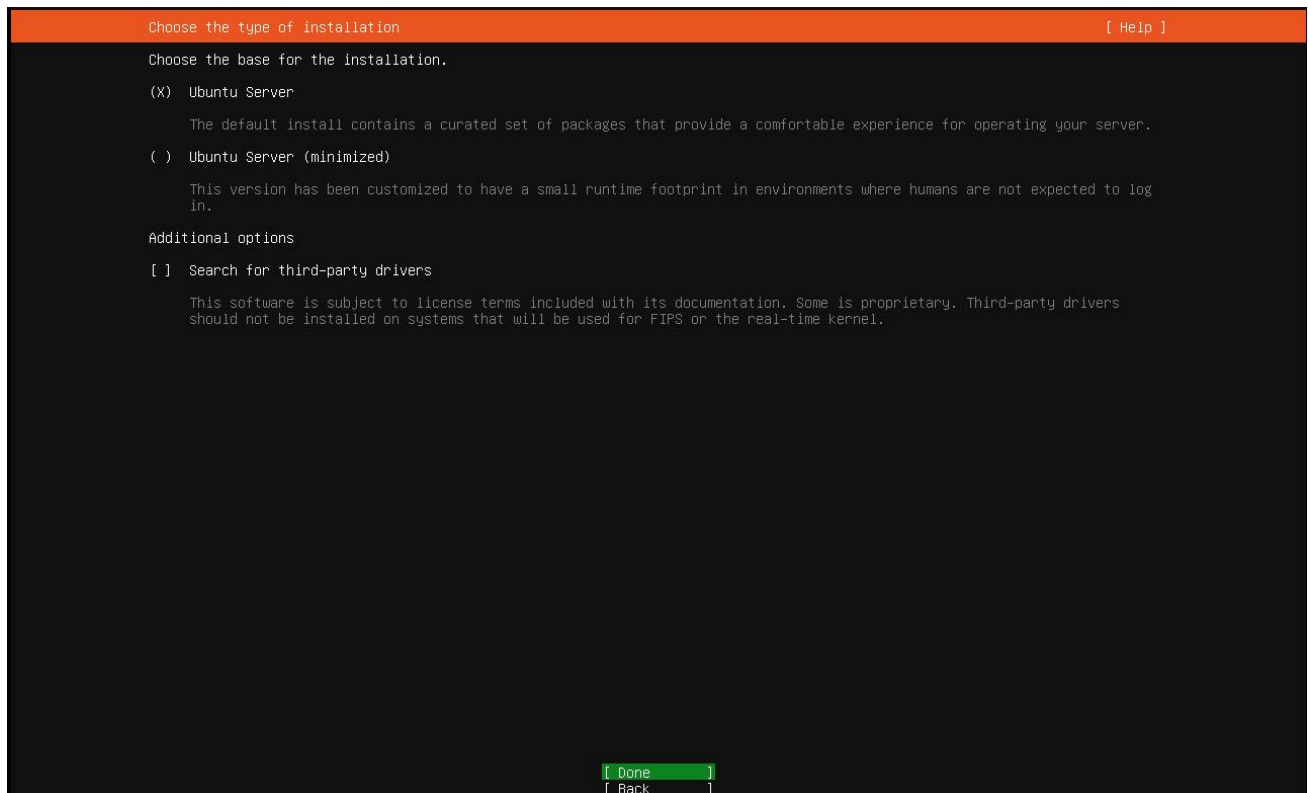


And then the keyboard variant is selected which is same as keyboard layout.



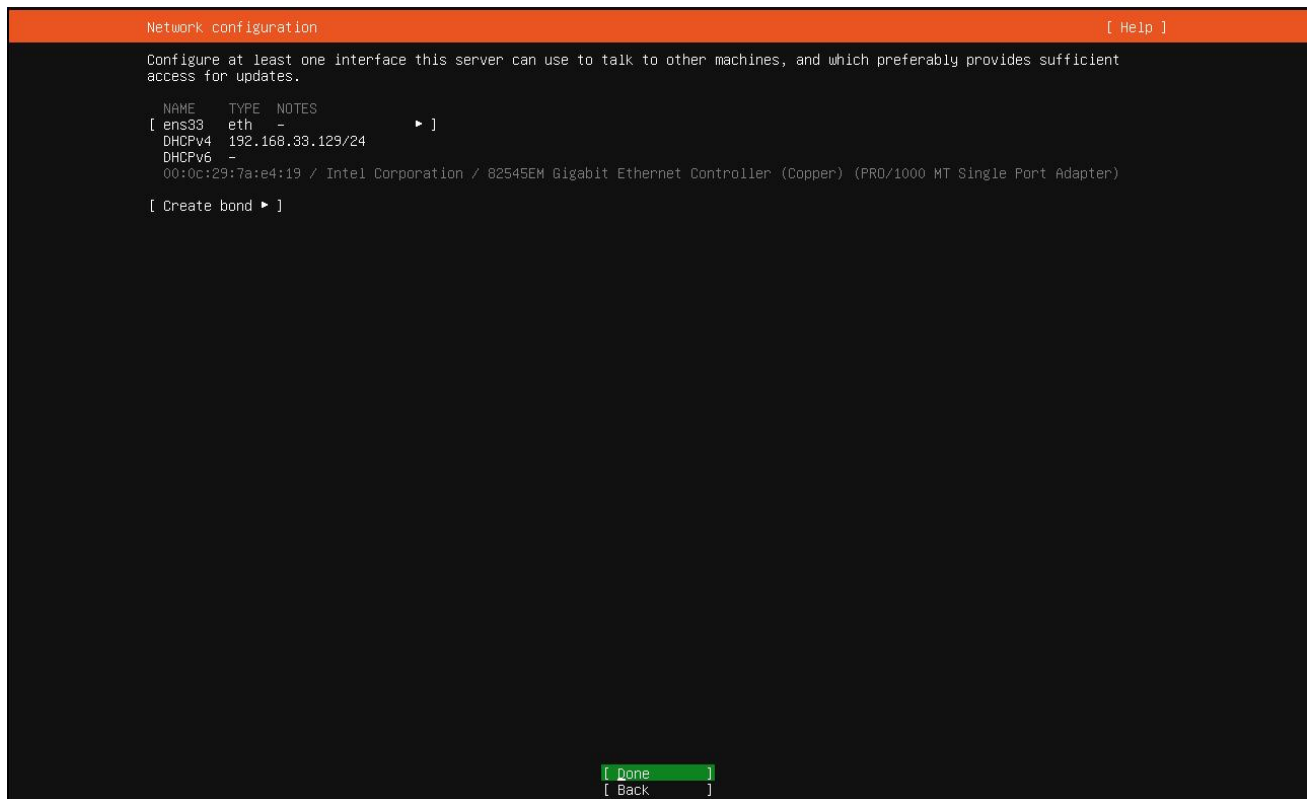
This image shows both: the selected keyboard layout as well as variant.

Task 07: Choose the Ubuntu Server Installation Type



Here the type of installation is selected for which Ubuntu Server is selected as base.

Task 08: Configure the Network

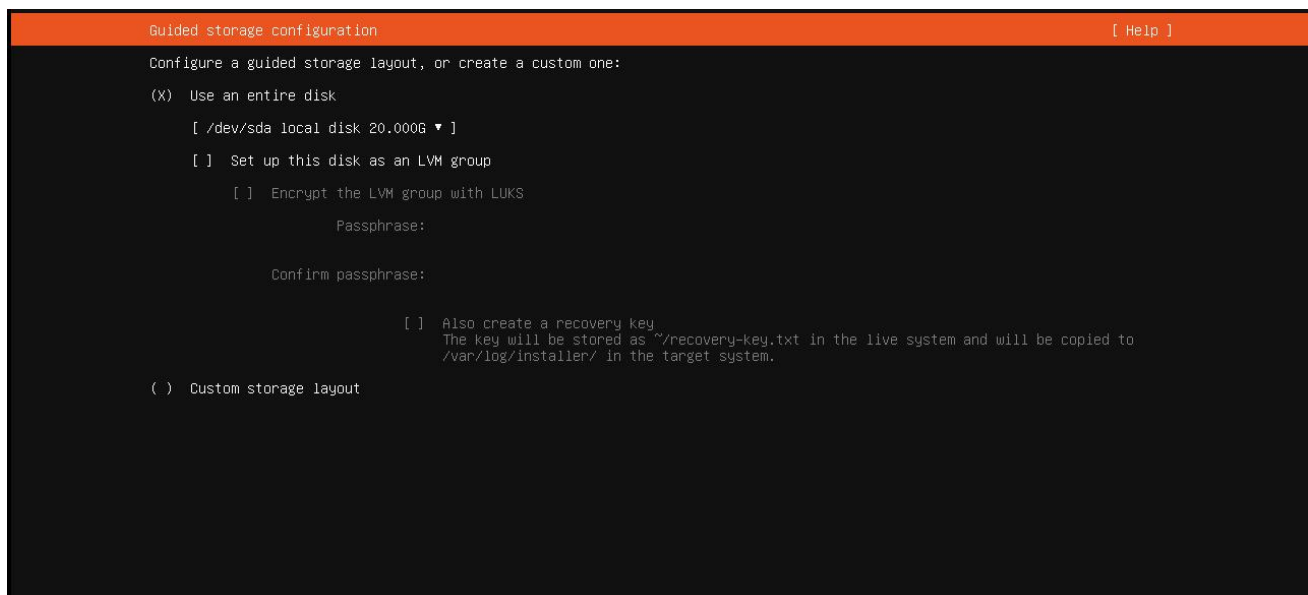


Here we can see the name of Network Interface that is detected by the Ubuntu Server Installer.

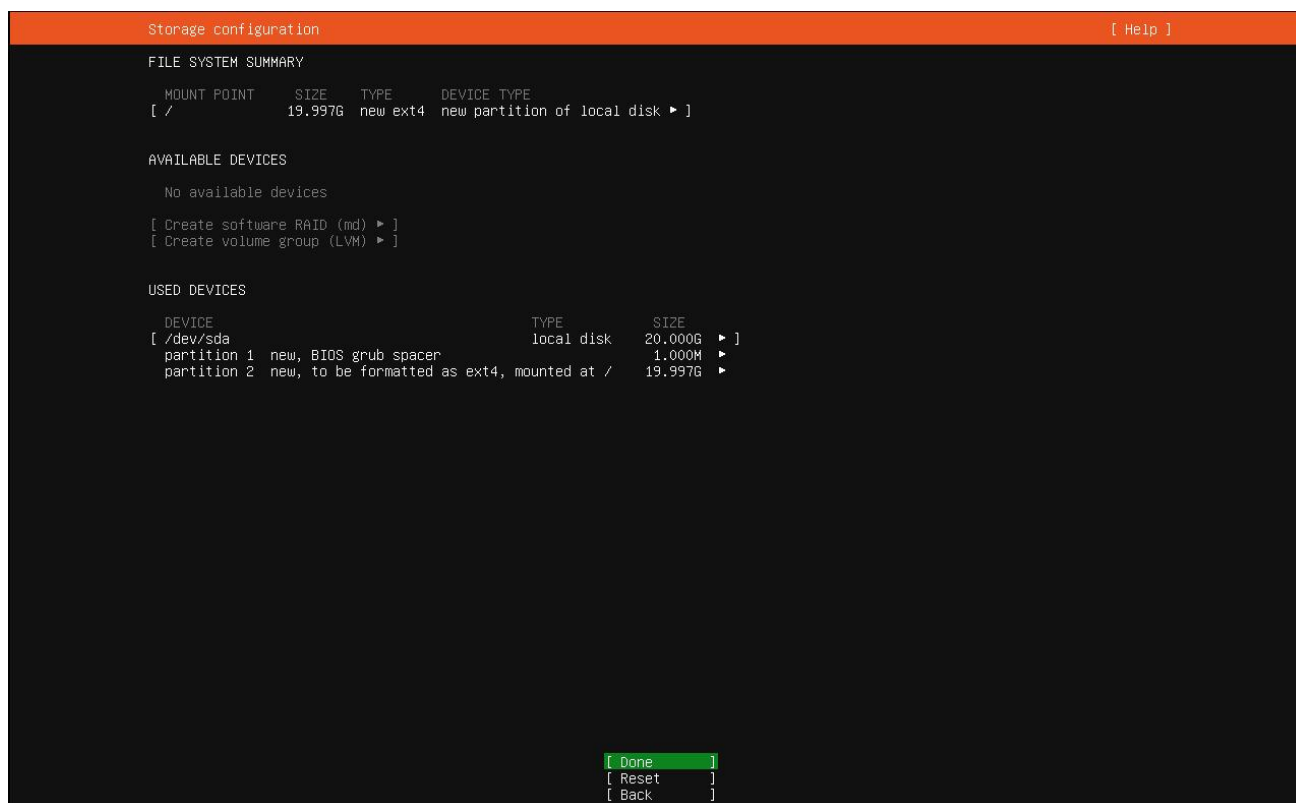


The Interface has successfully received the IP address automatically via DHCP.

Task 09: Configure Storage



Here “Use an entire disk” option is selected for guided storage configuration.



The complete storage configuration can be reviewed through this screen.

Task 10: Select the Installation Disk

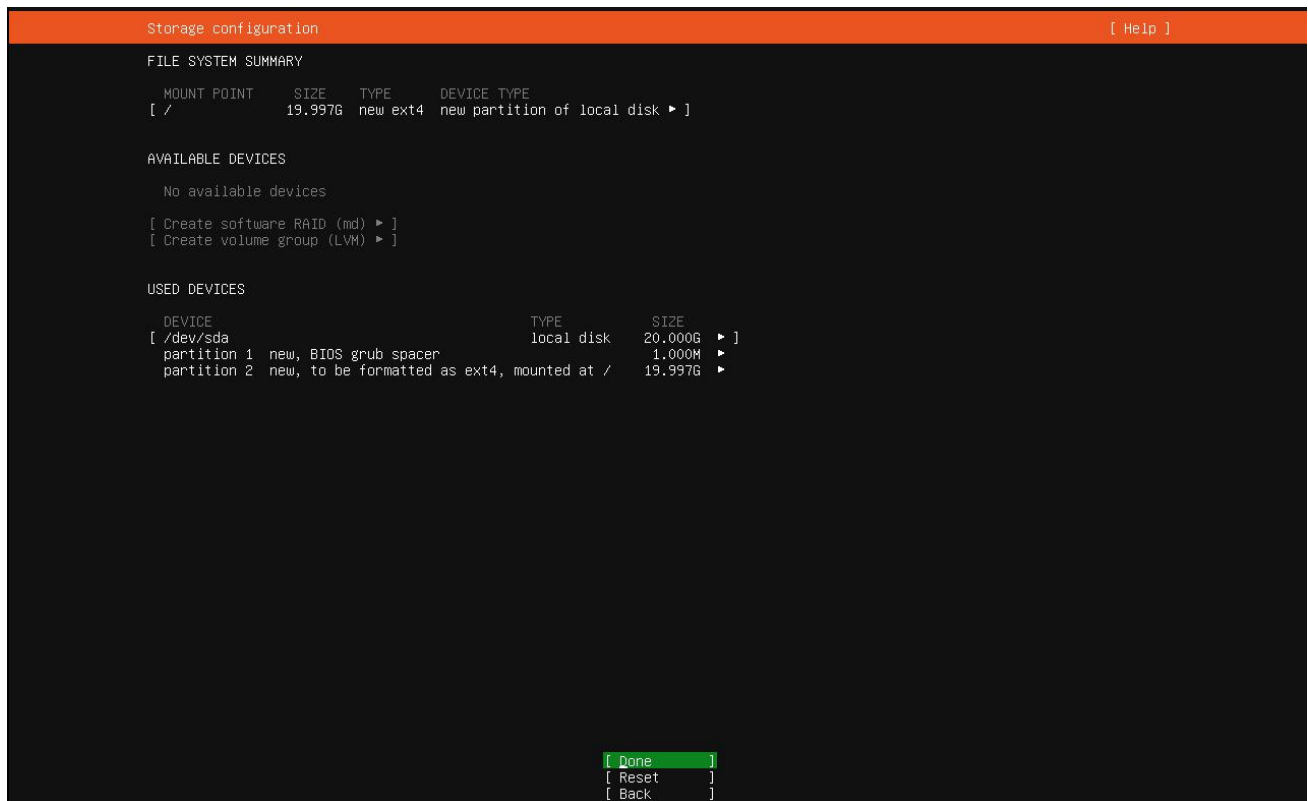
DEVICE	TYPE	SIZE
[/dev/sda	local disk	20.000G ▶]

This is the selected VMware Virtual disk.

USED DEVICES		
DEVICE	TYPE	SIZE
[/dev/sda	local disk	20.000G ▶]
partition 1	new, BIOS grub spacer	1.000M ▶
partition 2	new, to be formatted as ext4, mounted at /	19.997G ▶

These are the Virtual disk details including disk name and size.

Task 11: Review the Partitions

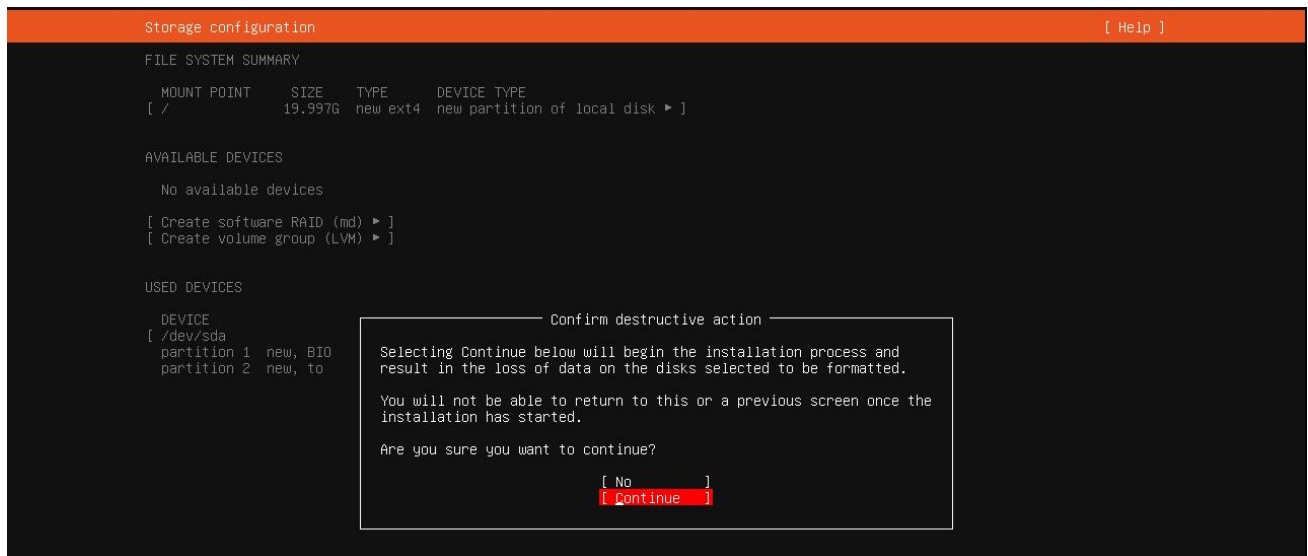


These are the disk partitions proposed by Ubuntu Server Installer.

Task 12: Confirm the Storage Changes



After reviewing the storage partitions, the final destructive-action warning appears.



After reading the warning and reviewing of virtual disk selection, the change are confirmed by pressing continue.

Task 13: Configure the Ubuntu Profile

Your name:

Actual name in Ubuntu Profile is entered.

Your servers name:
The name it uses when it talks to other computers.

The Server name is set to ubuntu.

Pick a username:

For the Ubuntu Profile, username is selected which is same as my GitHub username.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

Passwords are set here and the complete Ubuntu Profile is reviewed.

Task 14: Complete the Software Installation

Installing system

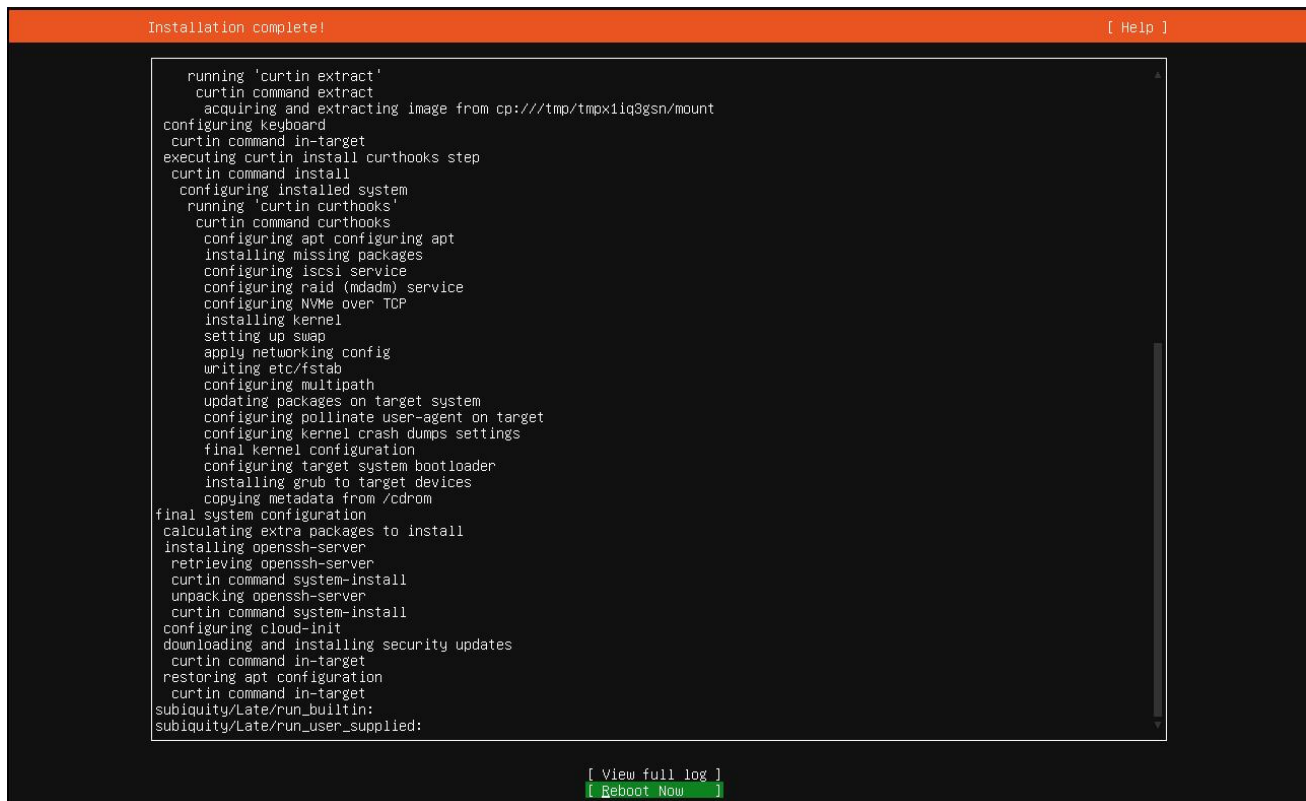
[Help]

```
subiquity/Userdata/apply_autoinstall_config:
subiquity/Package/apply_autoinstall_config:
subiquity/Debconf/apply_autoinstall_config:
subiquity/Kernel/apply_autoinstall_config:
subiquity/KernelCrashDumps/apply_autoinstall_config:
subiquity/Zdev/apply_autoinstall_config:
subiquity/Ad/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
  curtin command in-target
installing system
  executing curtin install initial step
  executing curtin install partitioning step
  curtin command install
    configuring storage
      running 'curtin block-meta simple'
      curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
      configuring partition: partition-0
      configuring partition: partition-1
      configuring format: format-0
      configuring mount: mount-0
  executing curtin install extract step
  curtin command install
    writing install sources to disk
      running 'curtin extract'
      curtin command extract
      acquiring and extracting image from cp:///tmp/tmpx1lq3gsn/mount
configuring keyboard
  curtin command in-target
  executing curtin install curthooks step
  curtin command install
    configuring installed system
      running 'curtin curthooks'
      curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel |
```

[View full log]

This is the installation progress of required software by Ubuntu Server.

Task 15: Reboot and Verify the Ubuntu Identity



After the installation completes, “Reboot Now” is selected.

```
ubuntu login: tuba-zainab
Password:
```

This is the Ubuntu login screen that appears after removing installation media and starting Ubuntu Server.

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sun Sep 13 07:34:43 AM UTC 2026

System load:  0.51               Processes:            239
Usage of /:   28.0% of 19.51GB   Users logged in:     0
Memory usage: 22%               IPv4 address for ens33: 192.168.83.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

tuba-zainab@ubuntu:~$ _
```

This welcome screen appears after signing in by using the username and password.

```
tuba-zainab@ubuntu:~$
```

The terminal prompt is followed by the exact required format.

i.e. <github-username>@ubuntu:~\$

Task 16: Find the Ubuntu Server IP Address

```
tuba-zainab@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:7a:e4:19 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamename enx000c297ae419
    inet 192.168.33.129/24 metric 100 brd 192.168.33.255 scope global dynamic ens33
        valid_lft 1462sec preferred_lft 1462sec
    inet6 fe80::20c:29ff:fe7a:e419/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

The “ip addr” command is used here that shows the IP address.

```
inet 192.168.33.129/24 metric 100 brd 192.168.33.255 scope global dynamic ens33
```

This screenshot clearly shows the IPv4 address.

Task 17: Connect to Ubuntu Server from Windows

```
C:\Users\tubaz>ssh tuba-zainab@192.168.33.129
The authenticity of host '192.168.33.129 (192.168.33.129)' can't be established.
ED25519 key fingerprint is SHA256:k0Mdn4sPeFwCOL621Qu956YXkUsv2wo7mDmSO/GPnf8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

SSH command is entered in Command Prompt which requires the server’s fingerprint to be accepted.

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.33.129' (ED25519) to the list of known hosts.
tuba-zainab@192.168.33.129's password:
```

After accepting fingerprint, it asks for ubuntu password.

```
tuba-zainab@192.168.33.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Sep 13 08:11:09 AM UTC 2026

System load:  0.0               Processes:    210
Usage of /:   28.1% of 19.51GB   Users logged in: 0
Memory usage: 23%              IPv4 address for ens33: 192.168.33.129
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Sep 13 07:47:25 2026 from 192.168.33.1
```

In this login screen, the Ubuntu password is entered which shows the successful SSH session.

```
tuba-zainab@ubuntu:~$
```

Remote SSH Prompt shows <github-username>@ubuntu:~\$

SSH Availability Proof:

The following screenshot clearly shows that SSH is already installed so we don't have to reinstall it.

```
tuba-zainab@ubuntu:~$ ssh -V
OpenSSH_10.2p1 Ubuntu-2ubuntu3.6, OpenSSL 3.5.5 27 Jan 2026
tuba-zainab@ubuntu:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
  Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
  Active: active (running) since Sun 2026-09-13 07:33:38 UTC; 32min ago
  Invocation: ea4590e9885740059752a6640261f527
  TriggeredBy: • ssh.socket
               Docs: man:sshd(8)
                   man:sshd_config(5)
  Main PID: 1202 (sshd)
  Tasks: 1 (limit: 1830)
  Memory: 3.1M (peak: 7.1M)
  CPU: 224ms
  CGroup: /system.slice/ssh.service
          └─1202 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```
